

From Coordinates to Compass: Euclid Meets Descartes

Problem Statement: *Given a hyperbola, construct its foci, transverse axis, conjugate axis, directrices, and asymptotes using only a compass and straightedge.*

To carry out the construction, we will use the following Euclidean techniques:

1. Bisecting a line segment
2. Drawing a perpendicular to a given line
3. Constructing tangents to a circle from an external point
4. Construction of a parallel line through a given point

Each construction will be accompanied by its respective proof, presented together with the steps.

1. Notation

For explanatory purposes, we will refer to the standard hyperbola to illustrate the objective and reasoning behind the construction. And will use these standard notations:

- > $a \equiv$ semi major axis of hyperbola
- > $e \equiv$ eccentricity
- > $b \equiv$ semi minor axis of hyperbola
- > $CA \equiv$ Conjugate axis
- > $TA \equiv$ Transverse axis
- > $S: \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0$
- > $T(x_1, y_1): \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1 = 0$
- > $S_1(x_1, y_1): \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1 = 0$

2. Finding the Centre

Diameter: A diameter is defined as the locus of the midpoints of a family of parallel chords of the hyperbola. Equivalently, it is a line through the centre such that all chords parallel to a given direction are bisected by it.

Eqn of chord with midpoint (x_1, y_1)

$$T(\alpha, \beta) = S_1(\alpha, \beta)$$

$$\frac{y\beta}{b^2} - \frac{x\alpha}{a^2} = \frac{\beta^2}{b^2} - \frac{\alpha^2}{a^2}$$

Let the parallel chord equation be : $y = mx + c$

After comparing the coefficient we get

$$\frac{\beta}{b^2} = \frac{\alpha}{ma^2}$$

Thus, getting the equation of diameter as

$$y = \frac{mb^2}{a^2}x \quad (1)$$

from eqn (2) we can conclude that the intersection of two diameter line will intersect on centre

Step 1: To determine the centre, we will construct two diameters and mark the intersection point.

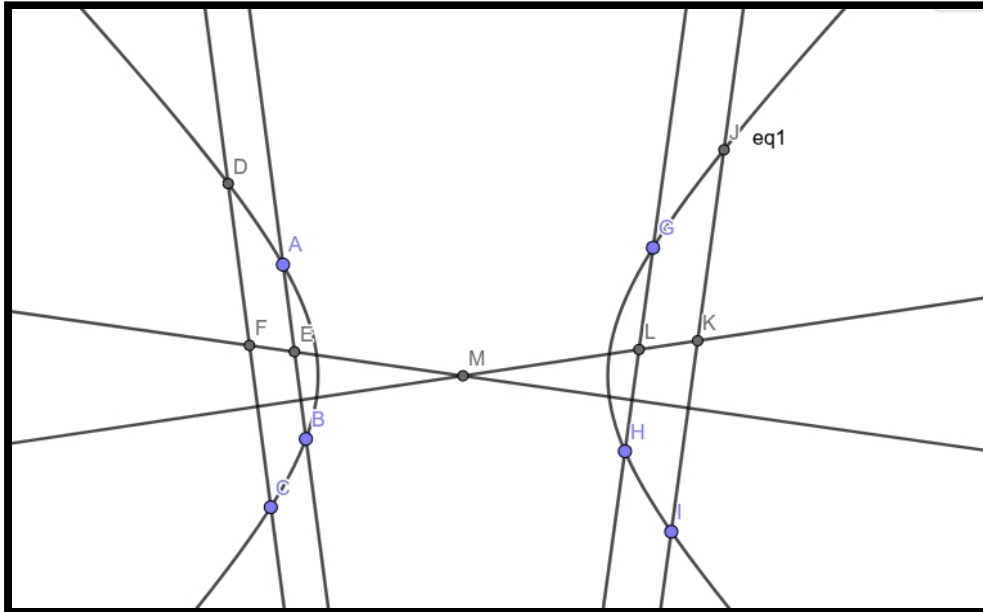


Figure 1: The intersection point M of the constructed diameters is the centre of hyperbola.

3. Constructing Transverse Axis(TA) and Conjugate Axis(CA)

Step 1: Draw a circle ω_1 centred at M cutting the hyperbola at N and O .

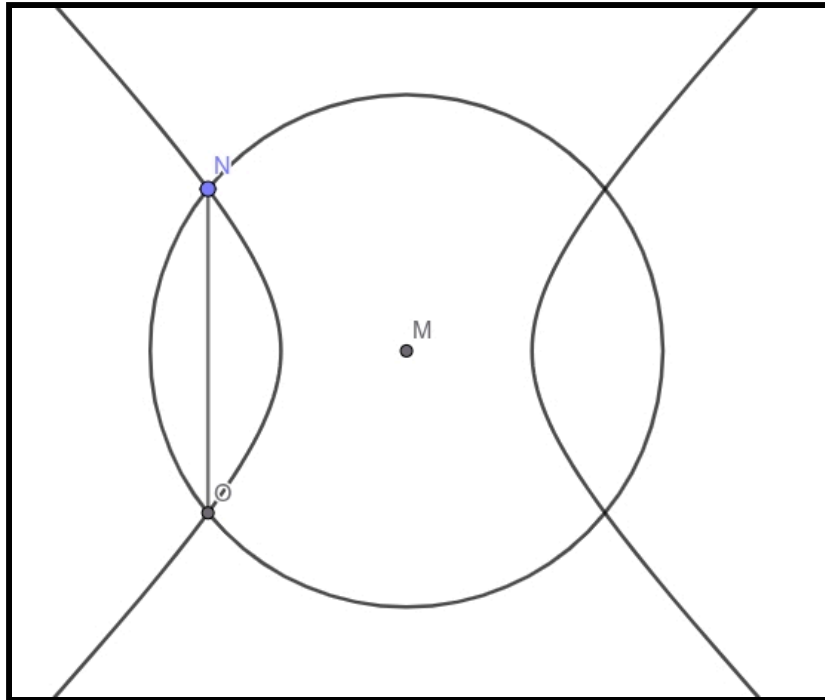


Figure 2: Marking the intersection of a circle with centre M and hyperbola

Step 2: Join M to the midpoint of NO to obtain the transverse axis. Through M , draw a perpendicular to it to obtain the conjugate axis.

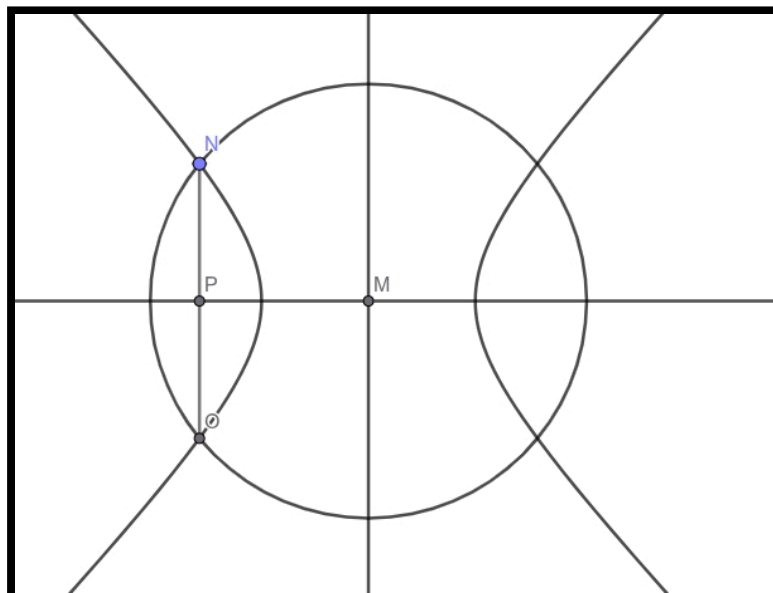


Figure 3: TA and CA is constructed

4. Motivation to step forward

Detour 1

Some Remarks from **cartesian geometry** that we will use to find the foci points and rest

$$1. e = \sqrt{1 + \frac{b^2}{a^2}}$$

2. **Vertices:** intersection between **TA** and **Hyperbola**

3. **Distance** between **vertices** = $2a \Rightarrow$ **distance** between **centre** and one **vertex** = a

4. **Focus** of hyperbola is at **distance** $ae = \sqrt{a^2 + b^2}$ from the centre on **TA**.

5. **Directrix** is the line at **distance** $\frac{a}{e}$ from **centre** and parallel to **CA**

6. The equation of **asymptotes** is $y = \pm \frac{b}{a}x$

7. **Distance** between **centre** and one of the **vertices** is a

Result 1

Finding x coordinate if the y coordinate is a (semi major axis)

$$S_1(h, a) : \frac{h^2}{a^2} - \frac{a^2}{b^2} - 1 = 0$$

$$h^2 = \left(1 + \frac{a^2}{b^2}\right)a^2 = \frac{a^4 e^2}{b^2}$$

$$h = \pm \frac{a^2 e}{b}$$

Conclusion: $\left(\pm \frac{a^2 e}{b}, a\right)$ will lie on hyperbola

Result 2

Observe this geometric figure; it follows directly from the Pythagorean theorem.

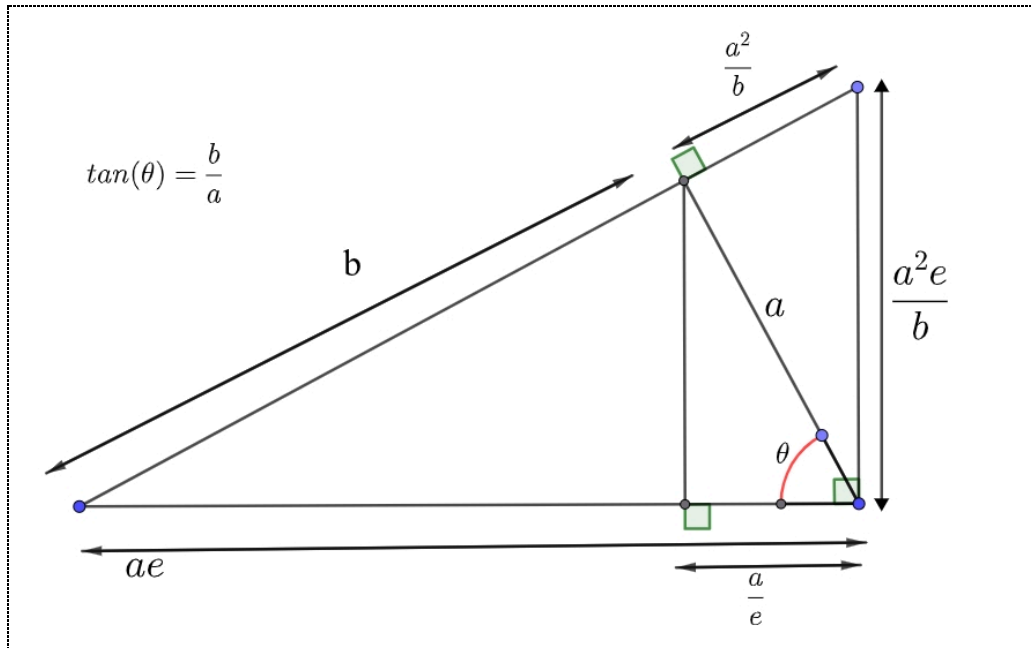


Figure 4: Triangle with dimensions mentioned

Figure 4 suggests the construction of the **above triangle**, which incorporates all the necessary elements (see **Detour 1**).

5. Construction of Foci, Asymptotes, Directrix and Asymptotes

Goal 1: to mark a point S from **centre** of hyperbola M on **CA** with $MS = \frac{a^2 e}{b}$

To accomplish **Goal 1**, we first construct a **circle** with **centre as M** and radius MV_1 , where V_1 is one of the **vertices** and this will lead to utilisation of **Result 1**.

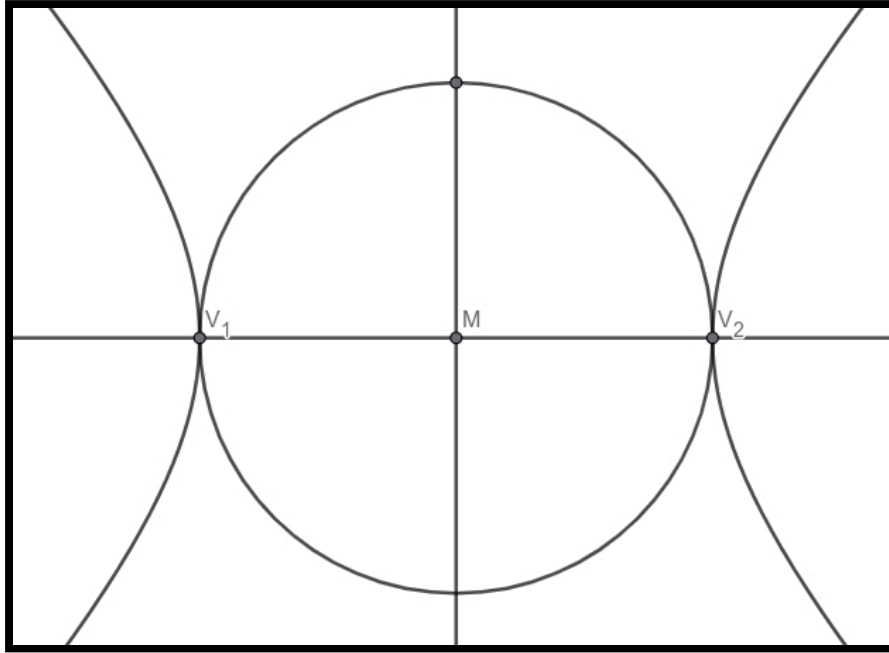


Figure 5: Construction of circle with radius a

We now construct a **line parallel** to the **TA**, passing through the circle and **CA**, This line will intersect the hyperbola at a point K , whose coordinates, using Result 1, are given

by $\left(\pm \frac{a^2 e}{b}, a \right)$.

After pointing K we will construct a circle ω_2 with centre M and radius as abscissa of K

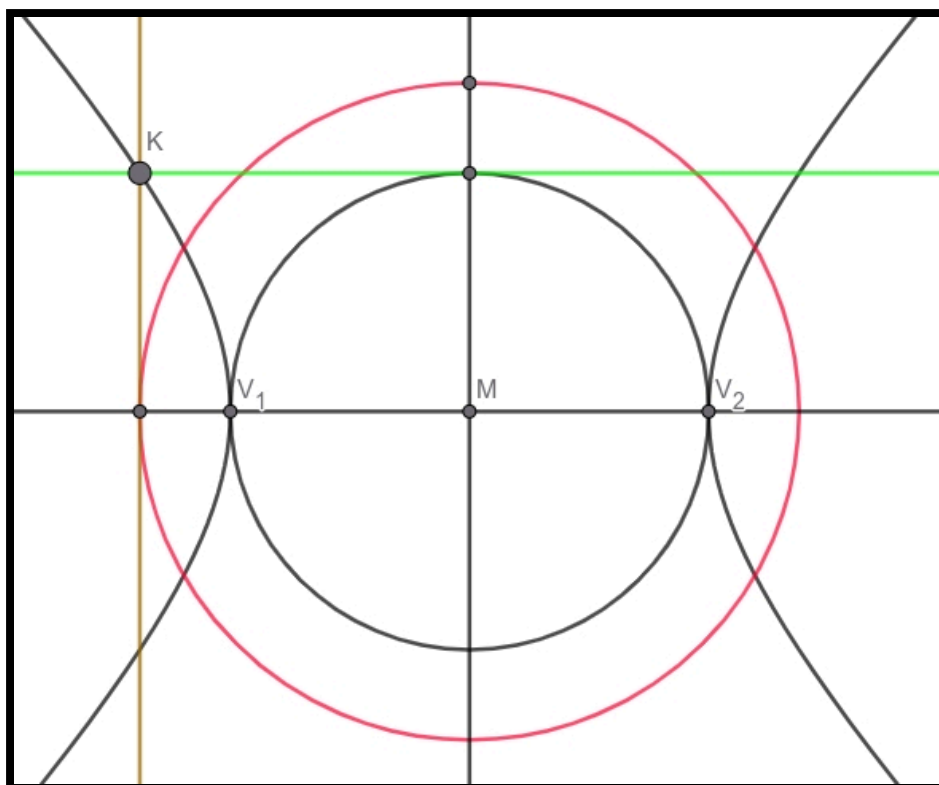


Figure 6: Construction of length $\frac{a^2e}{b}$ on **CA** using **Result 1**

Now you can easily see that if we make a **tangent** from **intersection** of ω_2 and **CA** (**which we will name as L**) to ω_1 (having radius a) will lead to the **triangle** needed to utilise **Result 2**

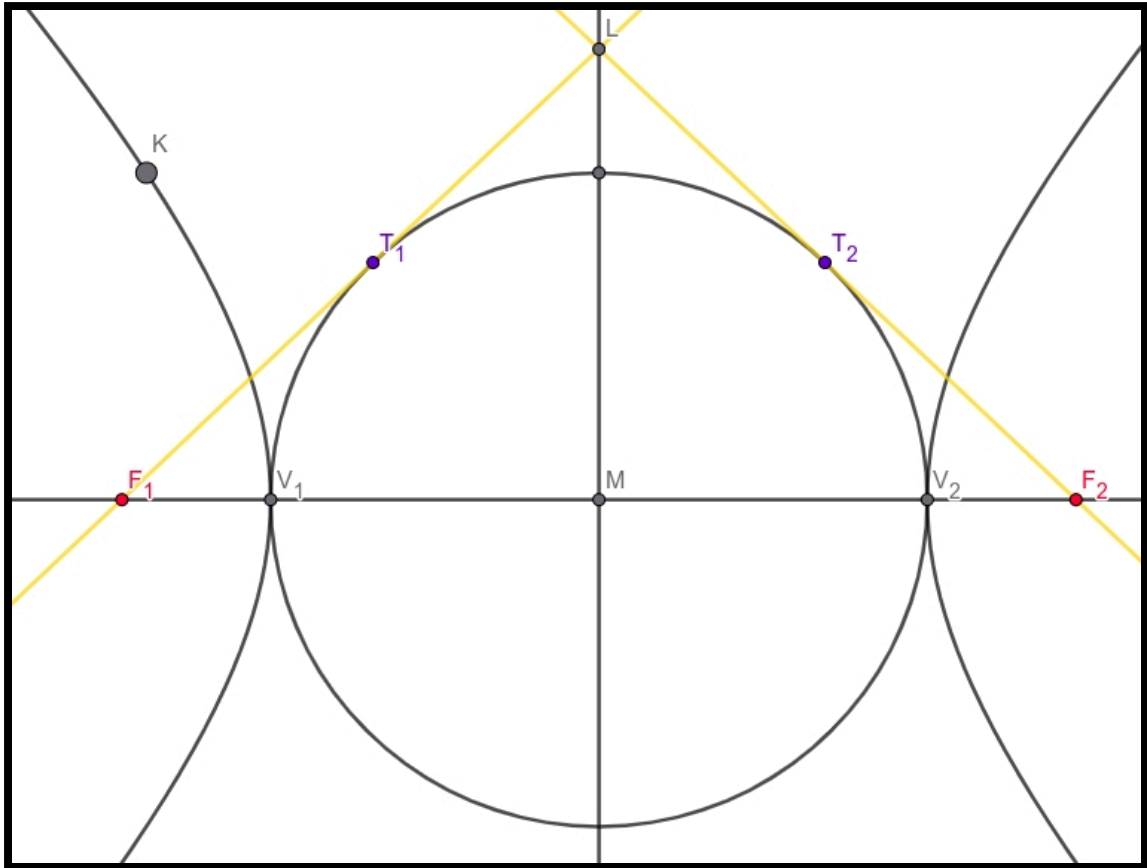


Figure 7: Yellow Lines - tangent to ω_1 from L , T_1 and T_2 are the touching points

From **Result 2** and **Figure 7**, it follows that F_1 and F_2 are the **Foci of the hyperbola**. The lines MT_1 and MT_2 will represent **asymptotes**. Furthermore, line parallel to **CA** and passing through T_1 or T_2 will be directrices

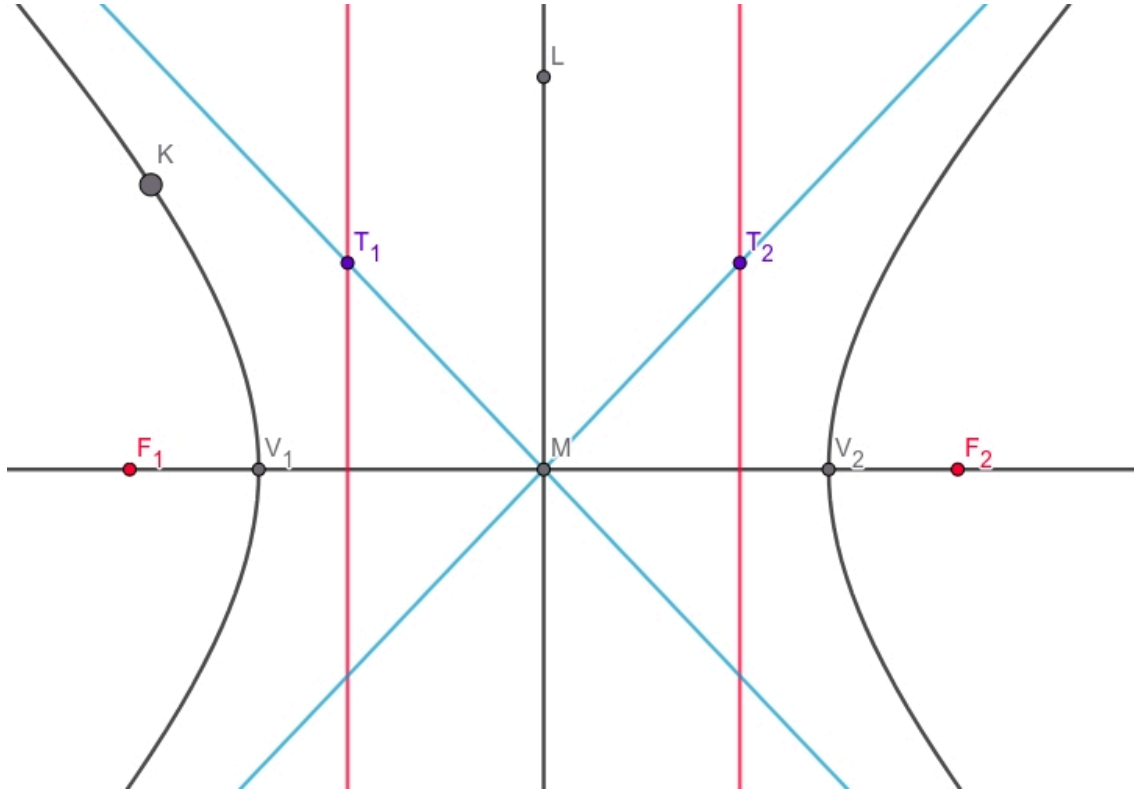


Figure 8: Concluding all the methods and results

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